

Memo 02: Reconstructed Delaware River Test Scenarios

Codex review for FVCOM-MP test-case recovery

April 27, 2026

1 Purpose

This memo reconstructs the intent of the three Delaware River test cases stored in `FVCOM_MP_test_run`: cases **a**, **b**, and **c**. The goal is to recover what is shared across the setups, what actually differs among them, and what those differences imply for later mass-conservation testing of the flocculation-enabled plastic model.

2 Recovered Test Bundle

2.1 Shared Domain and Forcing Files

The common input bundle in `FVCOM_MP_test_run/INPUT` defines a single Delaware River Estuary configuration with:

- 68416 mesh nodes and 30 sigma layers from `waterPACT_DRE_its.nc`;
- 100 open-boundary nodes in `waterPACT_obc_2019.nc` and `waterPACT_tsobc_2019.nc`;
- 9 river entries in `waterPACT_riv_floc_MP.nml`: 6 Delaware River points (`DR_1`–`DR_6`) and 3 Schuylkill River points (`SR_1`–`SR_3`);
- hourly wind and pressure forcing for calendar year 2019;
- a river forcing file that spans 2019/01/01 00:00:01 to 2020/01/01 00:00:01.

The model runs themselves are set from modified Julian day 58485.0 to 58848.0, i.e. 2019-01-02 00:00:00 through 2019-12-31 00:00:00. Therefore the shared forcing files provide roughly a one-day pad before the model start.

2.2 River Source Characteristics

The river forcing file `waterPACT_riv_floc_MP.nc` contains:

- `river_flux` in $\text{m}^3 \text{s}^{-1}$;
- `coarse_sand_1` in g L^{-1} ;
- `mp1` in kg m^{-3} .

Two points are especially clear:

1. `mp1` is constant in time and identical for all 9 rivers:

$$\text{mp1} = 1.754906975 \times 10^{-6} \text{ kg m}^{-3}.$$

2. `coarse_sand_1` varies in time, with domain-wide minimum and maximum values

$$4.516 \times 10^{-4} \text{ g L}^{-1} \leq \text{coarse_sand_1} \leq 1.6097 \times 10^{-2} \text{ g L}^{-1}.$$

Because the plastic concentration is constant, the imposed plastic source differs among rivers only through the discharge `river_flux`. Using the hourly records in `Times`, the implied annual river-source totals are approximately:

- Delaware group: $1.379 \times 10^{10} \text{ m}^3$ water, $2.420 \times 10^4 \text{ kg}$ plastic, $6.578 \times 10^7 \text{ kg}$ sediment;
- Schuylkill group: $3.915 \times 10^9 \text{ m}^3$ water, $6.870 \times 10^3 \text{ kg}$ plastic, $1.351 \times 10^7 \text{ kg}$ sediment.

3 Case-by-Case Reconstruction

Case	Intended role recovered from files	Plastic floc switch	Sediment setting	Other notable differences
a	Baseline plastic transport case without active plastic-sediment floc interaction.	<code>PLAST_FLOC = F</code> in <code>generic_plastic_and_sediment</code>	<code>SEDIMENT_MODEL = F</code> . The <code>sediment</code> file name is still present in the namelist but should be inactive.	Uses <code>STARTUP_TS_TYPE = 'observed'</code> from <code>waterPACT_DRE_its.nc</code> . NetCDF and restart output begin on MJD 58490.0, i.e. 5 days after the model start.
b	Main floc-enabled Delaware test with active suspended sediment and river plastic input.	<code>PLAST_FLOC = T</code> in <code>generic_plastic_and_sediment</code>	<code>SEDIMENT_MODEL = T</code> with <code>generic_sediment</code> as the only sediment class is <code>coarse_sand_1</code> with <code>SED_SD50 = 0.1 mm</code> .	Now uses <code>STARTUP_TS_TYPE = 'observed'</code> from <code>waterPACT_DRE_its.nc</code> , matching case a. NetCDF and restart output begin at the model start.

Case	Intended role recovered from files	Plastic floc switch	Sediment setting	Other notable differences
c	Sensitivity test relative to case b, most likely aimed at changing sediment character while keeping the plastic settings fixed.	PLAST_FLOC = T. <code>generic_plastic_c</code> is byte-for-byte identical to <code>generic_plastic_b</code>	SEDIMENT_MODEL = <code>ifp</code> with <code>generic_sediment_c</code> . The only actual change is SED_SD50 = 1.0 mm instead of 0.1 mm.	Same run configuration as case b except for output directory, sediment file name, plastic file name, and a shorter batch request on the cluster script. It now also uses <code>STARTUP_TS_TYPE = 'observed'</code> .

4 Important Non-Obvious Differences

4.1 The floc interaction term is explicitly sediment-size dependent

The water-column floc interaction is built from two rate fields in `mod_plast.F`: λ_a for aggregation and λ_d for disaggregation. These are recomputed every step when `PLAST_FLOC = T`. The sediment median diameter `sed(ised)%Sd50` enters the interaction in several direct ways:

- In `Calc_floc_MP_Aggregation_Rate`, each sediment class defines a floc-size gate d_{MP}^{max} from `Sd50`. If the plastic size exceeds this gate, that sediment class contributes zero aggregation.
- The same routine uses `Sd50` to define the sediment particle radius r_{sed} , which enters both the turbulent shear kernel $(r_{sed} + r_{plast})^3$ and the differential-settling kernel $(r_{sed} + r_{plast})^2 |w_{sed} - w_{plast}|$.
- The sediment mass concentration is converted to a number density using a spherical-particle mass $\propto r_{sed}^3$, so larger `Sd50` means fewer particles for the same sediment mass concentration.
- In `Calc_floc_MP_Disaggregation_Rate`, the maximum active sediment size across classes is used to rebuild the same d_{MP}^{max} gate, which then appears in the disaggregation coefficient.

4.2 Current consequence for the Delaware test cases

For the present Delaware inputs, the plastic particle size is approximately 0.100 mm = 100 μ m. The sediment-size gate therefore behaves very differently in cases b and c:

- Case b: `SED_SD50 = 0.1 mm = 100  $\mu$ m`. The implemented gate gives

$$d_{MP}^{max} = -0.0002 d_{floc}^2 + 0.36 d_{floc} \approx 34 \mu\text{m},$$

so a 100 μ m microplastic is larger than the permitted size. In practice this means the aggregation kernel is strongly size-excluded.

- Case c: `SED_SD50` = 1.0 mm = 1000 μm . The code then uses the large-floc cap

$$d_{MP}^{max} = 162 \mu\text{m},$$

so the same 100 μm microplastic is now allowed to aggregate.

This is not a small secondary effect. It means case **b** may behave close to a no-aggregation limit, while case **c** is the first case in this bundle that can genuinely exercise the floc aggregation kinetics for the chosen 100 μm plastic.

4.3 Aggregation changes the settling branch as well as the split state

The same code path assigns different settling velocities to the two plastic sub-states:

- `conc_d` uses `Calc_Wset_Plastic`, which for the present $\rho_p = 950 \text{ kg m}^{-3}$ plastic in seawater gives a negative settling velocity (upward or buoyant motion);
- `conc_a` is settled with the sediment settling velocity from the sediment model (`sed(1)%Wset` in the current implementation).

Therefore the floc interaction does not merely repartition `conc_a` and `conc_d`; once aggregation is active it can switch plastic from a buoyant branch to a downward-settling branch.

4.4 The startup T/S mismatch has now been removed

The run namelists for cases **b** and **c** have now been updated so that all three cases use `STARTUP_TS_TYPE` = 'observed' with the same startup file `waterPACT_DRE_its.nc`. The startup file contains non-trivial vertical structure:

- salinity `ssl` ranges from about -0.0028 to 35.19 PSU;
- temperature `tsl` ranges from about 4.82 to 13.43°C .

This removes one physical confounder from the original comparison between cases **a**, **b**, and **c**. A run-control difference still remains in the output timing:

- `RST_FIRST_OUT`: day 58490.0 in **a**, day 58485.0 in **b/c**;
- `NC_FIRST_OUT`: day 58490.0 in **a**, day 58485.0 in **b/c**.

4.5 Case c only changes sediment grain size, not the plastic setup

The `generic_plastic_b.inp` and `generic_plastic_c.inp` files are identical. The only content difference recovered between **b** and **c** is:

`SED_SD50` : 0.1 mm $\rightarrow$ 1.0 mm.

Other sediment parameters such as `SED_WSET`, `SED_TAUE`, and `SED_TAUD` remain unchanged in the provided files. So case **c** is best interpreted as a grain-size sensitivity test, not a broader sediment physics retuning.

4.6 No preserved model outputs are available in the case output folders

The directories `OUTPUT_a`, `OUTPUT_b`, and `OUTPUT_c` are presently empty. That means the original scenario intent has to be inferred entirely from the inputs and batch scripts; there is no saved output available here to confirm what diagnostics were originally examined.

5 Case b Output NetCDF Structure

The first successful case-b output stack inspected on Kestrel is:

```
/kfs3/scratch/yhuang168/waterPACT_MP_floc/OUTPUT_b/waterPACT_b_0001.nc.
```

It is a 64-bit NetCDF file with 160 records on the unlimited `time` dimension. A fuller script-oriented inventory has been saved separately as `analysis_notes/waterPACT_netcdf_structure_b.md`. The no-floc baseline structure for experiment `a` is recorded in `analysis_notes/waterPACT_netcdf_structure_a.md`.

5.1 Core Dimensions

Dimension	Size
<code>nele</code>	126722
<code>node</code>	68416
<code>siglay</code>	30
<code>siglev</code>	31
<code>three</code>	3
<code>time</code>	160, unlimited
<code>maxnode</code>	11
<code>maxelem</code>	9
<code>four</code>	4

5.2 Diagnostic-Relevant Variables

The fields most relevant for later mass-budget scripts are:

- geometry and volumes: `art1(node)`, `h(node)`, `zeta(node,time)`, `siglay(node,siglay)`, and `siglev(node,siglev)`;
- sediment forcing/state: `coarse_sand_1(node,siglay,time)` and `coarse_sand_1_bedfrac(node,time)`;
- total suspended plastic: `mp1(node,siglay,time)`;
- floc split: `mp1_agg(node,siglay,time)` and `mp1_dis(node,siglay,time)`;
- interaction rates and settling: `lambda_a_mp1`, `lambda_d_mp1`, `settle_vel_mp1`, and `settle_vel_floc_mp1`;
- bed exchange and bed inventory: `dep_flux_mp1(node,time)`, `ero_flux_mp1(node,time)`, `bot_mass_mp1(node,time)`, `bot_mass_agg_mp1(node,time)`, and `bot_mass_dis_mp1(node,time)`;
- supporting stress/density fields: `taub_total(node,time)`, `bottom_stress(node,time)`, `tau_ce_mp1(node,t`, and `ambient_rho(node,siglay,time)`.

For script design, the main consistency check is:

$$\text{mp1} \approx \text{mp1_agg} + \text{mp1_dis}.$$

Water-column inventory should use node-control-volume areas, likely `art1`, together with time-varying water depth and sigma-layer thickness. Bed inventory should use area integration of `bot_mass_*` fields. For the no-floc baseline case `a`, floc-only output variables may be absent after the NetCDF-registration fix because `PLAST_FLOC = F`.

6 Expected Mass-Conserving Signatures

If the model is internally mass conservative, the comparison among `a`, `b`, and `c` should look like a redistribution problem rather than a creation/loss problem.

6.1 Closed-system expectation

In a closed or internally isolated test, the correct quantity to compare is the total plastic inventory

$$M_{tot} = M_{water} + M_{bed}.$$

For all three cases, M_{tot} should remain constant up to numerical truncation. The floc interaction should only move plastic among:

1. disaggregated water-column plastic `conc_d`,
2. aggregated water-column plastic `conc_a`,
3. bottom or bed plastic mass.

6.2 What case a versus case b should look like

Because the current size gate in case `b` appears to exclude aggregation for a 100 μm microplastic in 100 μm sediment, case `b` should be close to the no-floc baseline in its floc kinetics:

- `conc_a` should remain near zero or at least much smaller than `conc_d`;
- the total plastic inventory in case `b` should closely track case `a` when summed over water plus bed;
- if case `b` shows a large total-mass deficit relative to case `a`, that would be more consistent with a numerical leak than with the intended floc physics.

6.3 What case a versus case c should look like

Case `c` is the one that should reveal the sediment-size sensitivity most clearly:

- `conc_a` should become nonzero where suspended sediment is present;
- `conc_d` should decrease where aggregation is active;
- the water-column plastic inventory can drop relative to case `a`, but that drop should be matched by an increase in bed plastic mass rather than by a loss of total plastic;
- because aggregated plastic uses the sediment settling branch while disaggregated plastic is buoyant, case `c` should show a stronger downward redistribution than case `a`.

6.4 What case b versus case c should look like

If the flocculation model is working and mass is conserved:

- case c should have more `conc_a` and more bed plastic than case b;
- case c should have less floating or disaggregated plastic than case b;
- the difference should be mainly in partitioning and vertical redistribution, not in total M_{tot} .

6.5 Open-boundary caveat

With realistic river and open-boundary forcing active, domain-integrated water-column mass does not need to match among cases at every output time because aggregation can change residence time and export. Even so, an internally conservative solution should still satisfy:

- no unexplained jump in `conc_a + conc_d` across the kinetics step itself;
- no persistent loss of $M_{water} + M_{bed}$ that cannot be explained by diagnosed export through open boundaries.

7 Supporting Scripts

Reusable inspection utilities have been placed in `FVCOM_MP_test_run/PYTHON`. The main script `inspect_delaware_cases.py` writes:

- `FVCOM_MP_test_run/PYTHON/output/case_inventory.json`;
- `FVCOM_MP_test_run/PYTHON/output/case_inventory.md`.

These files provide a machine-readable and human-readable summary of the same recovered scenario information.

8 Bottom Line

The recovered intent is most consistent with:

- **a**: no-floc baseline;
- **b**: floc-enabled reference case;
- **c**: floc-enabled sediment-size sensitivity case.

The most important code-level finding is that the flocculation interaction depends directly on sediment size, and with the present inputs that dependence likely suppresses aggregation in case **b** while allowing it in case **c**. After aligning startup T/S across all three run namelists, the main physical comparison to watch is whether case **c** shows redistribution into `conc_a` and the bed without losing total plastic mass.